NEXT GEN SOFTWARE HUB

CYBER SECURITY INTERNSHIP

WEEK 01 TASK 01 REPORT

Cybersecurity Basics

Understanding the CIA Triad, Threats & Vulnerabilities, and Types of Cyber Attacks

TOPICS

- CIA Triad
- Threats & Vulnerabilities
- Types of Cyber Attacks

TOOLS

- N/A

RESOURCES

- Intro to Cybersecurity - Cisco
- Cybrary's Cybersecurity Overview

TASK

- Write a blog/post explaining the CIA Triad with real-world examples.

PHASE 1: FOUNDATIONS (WEEK 1–4)

- Week 1: Cybersecurity Basic



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Introduction to Cybersecurity

In today's digital world, computers, smartphones, networks, and online services play an important role in everyday life. People use digital platforms to communicate, access banking services, attend classes, conduct business, and store personal information. Organizations also depend on computer systems to manage customer records, financial transactions, employee information, and other critical operations. Although these technologies offer convenience and efficiency, they also create security risks when information and systems are not adequately protected.

Cybersecurity is the practice of protecting computer systems, networks, applications, and data against unauthorized access, misuse, disruption, modification, and destruction. Its purpose is not only to prevent cybercriminals from accessing information but also to ensure that information remains accurate and that essential systems continue to operate when needed.

One of the most important concepts in cybersecurity is the CIA Triad, which stands for Confidentiality, Integrity, and Availability. These three principles provide a fundamental framework for understanding information security. They help individuals and organizations identify what needs protection, recognize the possible consequences of security incidents, and select appropriate security controls.

For example, a bank must protect its customers' financial information from unauthorized access, ensure that account balances and transactions remain accurate, and keep its banking services accessible to legitimate customers. These responsibilities represent confidentiality, integrity, and availability.

This blog post explains the CIA Triad, provides practical real-world examples, explores common cybersecurity threats and vulnerabilities, and examines how different types of cyberattacks can compromise information security.

1. What Is the CIA Triad?

The CIA Triad is a foundational information security model consisting of three essential principles: Confidentiality, Integrity, and Availability. Each principle addresses a different security objective, but all three work together to protect information and information systems.

The model can be applied to many environments, including personal computers, corporate networks, university databases, hospital information systems, cloud platforms, and online banking applications.

The three principles can be summarized as follows:

- **Confidentiality:** Ensuring that information is accessible only to authorized individuals and systems.
- **Integrity:** Ensuring that information remains accurate, complete, and protected against unauthorized changes.
- **Availability:** Ensuring that authorized users can access information and services when they need them.

A security incident may affect one, two, or all three principles simultaneously. For this reason, cybersecurity professionals must understand how the principles relate to one another when evaluating risks and designing security measures.

2. Confidentiality: Protecting Information from Unauthorized Access

2.1 What is confidentiality?

Confidentiality is the principle of ensuring that sensitive information is disclosed only to authorized individuals, systems, or applications. It prevents unauthorized parties from viewing, obtaining, or exposing information that should remain private.

Information requiring confidentiality may include passwords, personal identification details, financial records, medical histories, business strategies, examination records, and confidential communications.

For example, when a customer logs into an online banking application, the customer's account information should be available only to the account holder and other properly authorized parties. An unauthorized person should not be able to view the customer's balance, transaction history, or personal details.

Confidentiality is important because unauthorized disclosure can result in identity theft, financial losses, privacy violations, reputational damage, and other harmful consequences.

2.2 Real-world example: Online banking and phishing

Imagine that a customer receives an email claiming to be from their bank. The email warns that the customer's account will be suspended unless they verify their details immediately. It contains a link to a fraudulent website designed to resemble the bank's legitimate login page.

The customer enters their username and password, unknowingly sending these credentials to a cybercriminal. The attacker then uses the stolen credentials to access the customer's banking information.

In this situation, confidentiality has been compromised because private information has been disclosed to an unauthorized person.

The attack may also lead to financial fraud if the attacker uses the compromised account to conduct unauthorized transactions. However, the initial confidentiality failure occurs when the credentials and private account information are exposed.

This example demonstrates that cybersecurity risks do not always begin with sophisticated technical attacks. Sometimes, attackers exploit human trust and a lack of awareness to obtain sensitive information.

2.3 How can confidentiality be protected?

Organizations and individuals can protect confidentiality through several security controls.

- **Strong passwords and multi-factor authentication:** Users should create strong, unique passwords for their accounts. Multi-factor authentication (MFA) adds another verification step, such as an authenticator application or security key, making unauthorized access more difficult even if a password is stolen.
- **Encryption:** Encryption converts readable information into a protected form that cannot be easily understood without the appropriate decryption key. It helps protect sensitive data both when stored on a device and when transmitted across a network.
- **Access control:** Access control determines who can access specific information and what actions they are allowed to perform. For example, a university lecturer may be authorized to

update examination results for assigned courses, while other students should not have permission to modify those records.

- **User awareness and phishing prevention:** Users should learn to recognize suspicious links, unexpected attachments, fraudulent login pages, and messages requesting confidential information. Organizations should also establish procedures for reporting suspected phishing attempts.
- **Data minimization:** Organizations should collect and retain only the information they genuinely need. Reducing the amount of sensitive information stored can limit the potential impact of a security breach.

These measures help reduce the risk of unauthorized disclosure and maintain the privacy of sensitive information.

3. Integrity: Maintaining Accurate and Trustworthy Information

3.1 What is integrity?

Integrity is the principle of maintaining the accuracy, completeness, consistency, and trustworthiness of information throughout its lifecycle. It ensures that information is not modified, deleted, or manipulated without proper authorization.

Data integrity is essential because individuals and organizations rely on information to make decisions. If information is inaccurate or has been secretly altered, the decisions based on that information may also be incorrect.

For example, a bank must maintain accurate account balances and transaction records. A hospital must preserve the accuracy of patient records. A university must ensure that examination results reflect the marks legitimately awarded to students.

Integrity does not mean that information can never be changed. Authorized updates are often necessary. Instead, integrity requires that changes occur through approved processes and that unauthorized or accidental modifications can be prevented or detected.

3.2 Real-world example: University examination results

Imagine that a university stores students' examination marks in an electronic database. A student receives a mark of 45%, but an unauthorized person gains access to the system and changes the recorded mark to 85%.

If the change remains undetected, the database no longer represents the student's legitimate examination result. Academic decisions based on the altered information may become unfair or incorrect.

This is an integrity failure because information has been modified without authorization.

The same principle applies to financial systems. If an attacker changes the amount of a financial transaction or alters an organization's accounting records, the resulting information may misrepresent the actual financial position of the organization.

Integrity therefore protects the reliability of records and helps ensure that decisions are based on accurate information.

3.3 How can integrity be protected?

Several security controls can help preserve information integrity.

- **Access permissions:** Only authorized users should be permitted to modify sensitive information. Permissions should be assigned according to each user's responsibilities, following the principle of least privilege.
- **Cryptographic hashes:** A cryptographic hash function produces a fixed-length value from a file or other data. Comparing a newly calculated hash with a trusted previous hash can help identify whether the data has changed. However, the reference hash must itself be protected against unauthorized modification.
- **Audit logs:** Audit logs record relevant activities, such as logins, changes to records, and administrative actions. These records help investigators identify suspicious activity and determine what changes occurred and when.
- **Data validation:** Systems should validate information before accepting it. For example, a banking application can check that transaction amounts follow the expected format and that a transaction complies with relevant account rules.
- **Backups and recovery procedures**: Maintaining reliable backups can help restore correct information following accidental deletion, corruption, or malicious modification. Backups should be protected and periodically tested.
- **Change management:** Organizations should use approved procedures for making important system and data changes. This helps prevent unauthorized modifications and provides accountability.

Together, these controls help ensure that information remains trustworthy and that unauthorized changes can be prevented, detected, or corrected.

4. Availability: Ensuring Access to Information and Services

4.1 What is availability?

Availability is the principle of ensuring that information, applications, networks, and services remain accessible to authorized users when required.

A system may have strong access controls and accurate information, but it will still fail to meet users' needs if they cannot access it. Availability therefore focuses on reliable operations, minimizing unnecessary downtime, and restoring services after disruptions.

Availability is especially important for hospitals, banks, online businesses, educational institutions, communication providers, and public services.

For example, a customer may need to access online banking to make an urgent payment. If the banking service is unavailable because of a cyberattack or technical failure, the customer may be unable to complete the transaction when needed.

4.2 Real-world example: Ransomware affecting hospital systems

Imagine that a hospital's computer network is infected with ransomware. The malicious software encrypts important files and prevents medical staff from accessing some patient records and administrative systems.

As a result, staff may experience difficulties retrieving medical histories, checking appointment schedules, or accessing information needed for hospital operations.

The incident compromises availability because authorized users cannot access essential information when they need it. Depending on the circumstances, it may also affect confidentiality if the attackers steal patient information before encrypting the files.

This example demonstrates that availability is not simply a technical requirement. In critical environments such as healthcare, system downtime can disrupt important services and potentially place people at risk.

4.3 How can availability be protected?

Organizations can improve availability through several measures.

- **Regular backups**: Important information should be backed up regularly. Backups should be stored securely, separated from systems that could be compromised by the same attack, and tested to confirm that recovery works.
- **Redundancy:** Redundancy involves having additional systems, network connections, or components that can take over when a primary component fails. It can reduce the likelihood that a single failure will interrupt an entire service.
- **System maintenance and patch management:** Operating systems, applications, and network devices should be maintained and updated appropriately. Updates can address vulnerabilities that attackers might exploit to disrupt services.
- **Protection against denial-of-service attacks:** Organizations can use network monitoring, traffic filtering, rate limiting, and appropriate service-provider protections to reduce the impact of denial-of-service attacks.
- **Incident response and disaster recovery:** An incident response plan outlines how an organization identifies, contains, and responds to security incidents. A disaster recovery plan explains how important systems and services can be restored after a major disruption.
- **Monitoring and maintenance**: Monitoring systems can help administrators detect unusual activity, hardware failures, resource exhaustion, and other conditions that may threaten availability. Regular maintenance can help identify problems before they cause significant downtime.

These measures help organizations maintain reliable services and recover more effectively when incidents occur.

5. How the Three Principles Work Together

Confidentiality, Integrity, and Availability are separate concepts, but they are closely connected. Protecting one principle does not automatically guarantee that the other two are protected.

Consider an online banking system.

Confidentiality: Customers' financial information must be protected from unauthorized access.

Integrity: Account balances and transaction records must remain accurate and must not be changed without authorization.

Availability: Customers must be able to access banking services when required, subject to legitimate maintenance and unexpected disruptions.

Suppose a cybercriminal steals a customer's login credentials. This threatens confidentiality because private information may be exposed. If the attacker then changes account details or initiates unauthorized transactions, integrity may also be affected. If the attacker or resulting incident causes the banking service to become unavailable, availability is compromised as well.

This example illustrates how one security incident can affect several parts of the CIA Triad.

The model helps security professionals evaluate the potential impact of incidents, identify which assets require protection, and choose security measures that address the relevant risks.

6. Understanding Cybersecurity Threats and Vulnerabilities

To understand why the CIA Triad matters, it is necessary to understand the risks that can compromise information security.

6.1 What is a cybersecurity threat?

A cybersecurity threat is a potential cause of harm to information, systems, or networks. Threats may arise from malicious actors, human mistakes, software failures, or environmental events.

Examples include cybercriminals attempting to steal passwords, malicious software infecting computers, employees misusing their access privileges, and power failures disrupting computer systems.

A threat does not necessarily result in a successful attack. Harm occurs when circumstances allow the threat to affect an asset, often by exploiting a vulnerability.

6.2 What is a vulnerability?

A vulnerability is a weakness in a computer system, application, network, configuration, or security procedure that could be exploited or otherwise lead to a security problem.

Common examples include:

- Outdated software containing known security weaknesses.
- Weak or reused passwords.
- Incorrectly configured network devices or cloud services.
- Excessive user permissions.
- Applications containing security flaws.
- Inadequate backups or poorly tested recovery procedures.

Vulnerabilities increase risk because they may create opportunities for unauthorized access, data manipulation, or service disruption.

6.3 How threats and vulnerabilities are related

Consider an organization that uses outdated software containing a known security vulnerability.

The outdated software represents a weakness in the organization's environment. A cybercriminal seeking to gain unauthorized access represents a threat actor. If the criminal exploits the weakness to compromise the system, an attack has occurred.

The possible consequences depend on what the attacker achieves. Stolen information can compromise confidentiality, unauthorized changes can compromise integrity, and disrupted services can compromise availability.

Understanding this relationship helps organizations identify weaknesses, assess potential risks, and apply appropriate security controls before those weaknesses are exploited.

7. Common Types of Cyberattacks and Their Impact on the CIA Triad

Cyberattacks use different techniques to compromise information, gain unauthorized access, or disrupt systems. The following examples demonstrate how common attacks can affect the three principles of information security.

7.1 Phishing attacks

Phishing is a form of social engineering in which attackers use deceptive emails, messages, phone calls, or websites to trick people into revealing information or performing unsafe actions.

For example, an attacker may send a message pretending to be from a bank and direct the recipient to a fraudulent login page.

Primary CIA impact**:** Confidentiality, because credentials or sensitive information may be disclosed. If stolen credentials are used to change records or disable an account, integrity or availability may also be affected.

7.2 Malware attacks

Malware is malicious software designed to perform unauthorized or harmful activities. Examples include spyware, worms, trojans, and ransomware.

For example, spyware may collect sensitive information from an infected device, while a trojan may provide an attacker with unauthorized access.

Primary CIA impact: The impact depends on the malware's purpose. Spyware can compromise confidentiality, while malware that alters files can affect integrity or malware that disrupts system operation can affect availability.

7.3 Ransomware attacks

Ransomware is a type of malware that typically encrypts files or otherwise prevents access to systems until the attacker demands payment. Some ransomware campaigns also steal information and threaten to disclose it.

For example, ransomware affecting a business may prevent employees from accessing shared files, customer records, and essential applications.

Primary CIA impact: Availability, because information and systems may become inaccessible. Confidentiality may also be compromised if data is stolen, and integrity may be affected if files or records are altered or corrupted.

7.4 Denial-of-service attacks

A denial-of-service (DoS) attack attempts to make a system or service unavailable by overwhelming it with requests or exhausting its resources. A distributed denial-of-service (DDoS) attack uses multiple sources to generate the disruptive traffic.

For example, attackers may direct a large volume of traffic toward an online store, preventing legitimate customers from loading the website or completing purchases.

Primary CIA impact: Availability, because legitimate users may be unable to access the service.

7.5 Man-in-the-middle attacks

A man-in-the-middle (MitM) attack occurs when an attacker intercepts or manipulates communications between two parties. Depending on the circumstances and available protections, the attacker may observe sensitive information or modify messages in transit.

For example, if an attacker successfully intercepts inadequately protected communications between a user and an online service, the attacker may attempt to obtain sensitive information or alter transmitted data.

Primary CIA impact: Confidentiality may be compromised if information is exposed, while integrity may be compromised if communications are modified.

7.6 Password attacks

Password attacks involve attempts to obtain, guess, or misuse authentication credentials. Techniques include password guessing, credential stuffing using previously exposed passwords, and brute-force attempts against authentication systems.

For example, an attacker may use credentials leaked from one website to attempt to access a person's account on another service where the same password has been reused.

Primary CIA impact: Confidentiality may be compromised when unauthorized access exposes private information. Integrity and availability may also be affected if the attacker changes account information or locks the legitimate user out.

These examples demonstrate that cyberattacks can have different consequences depending on the target, the attack method, the protections in place, and the attacker's objectives.

8. Applying the CIA Triad in Everyday Life

The CIA Triad is not limited to large companies or advanced technical environments. It applies to everyday activities involving smartphones, email, social media, cloud storage, online banking, and educational platforms.

Consider a student who uses an online learning platform to access course materials, submit assignments, and view examination results.

Confidentiality: The student's account and personal information should be accessible only to the student and other authorized users. Strong authentication and appropriate access controls help protect this information.

Integrity: Submitted assignments, examination results, and academic records should remain accurate. Unauthorized individuals should not be able to change marks or modify submitted work.

Availability: The learning platform should remain accessible when students need to study, submit assignments, or access important academic information. Reliable infrastructure, maintenance, backups, and recovery procedures can help support availability.

If a student's account is compromised, all three principles may be affected. An attacker might read private information, change submitted work or account details, or prevent the student from accessing the account.

This example demonstrates how the CIA Triad can help individuals recognize everyday security risks and understand why protecting accounts and devices is important.

9. Why the CIA Triad Matters in Cybersecurity

The CIA Triad provides a common foundation for planning and evaluating information security. It helps organizations move beyond simply asking whether a system is protected and consider the specific security objectives that must be achieved.

Its importance can be explained in several ways.

First, it helps identify security requirements. Organizations can determine which information must remain confidential, which records require strong integrity protections, and which services must remain available.

Second, it supports risk assessment. Security professionals can examine how a particular threat or vulnerability might affect confidentiality, integrity, or availability and estimate the potential consequences.

Third, it guides the selection of security controls. Encryption and access controls can support confidentiality, audit logs and integrity checks can support integrity, and backups and redundant infrastructure can support availability.

Fourth, it supports incident response. When a security incident occurs, understanding which principles have been compromised helps responders assess the impact, prioritize recovery, and determine which corrective actions are needed.

Finally, it promotes security awareness. The CIA Triad gives students, employees, and technology users a straightforward way to understand the purpose of cybersecurity measures.

Although the model is fundamental, it is not a complete security programme by itself. Organizations must also consider other requirements, including authentication, accountability, privacy, legal obligations, and business continuity.

Conclusion

The CIA Triad is one of the fundamental models in cybersecurity because it explains the three primary objectives of protecting information: Confidentiality, Integrity, and Availability.

Confidentiality ensures that information is not disclosed to unauthorized parties. Integrity ensures that information remains accurate, complete, and protected against unauthorized modification. Availability ensures that authorized users can access information and services when they need them.

The examples discussed in this blog demonstrate how these principles apply in practical situations. A phishing attack can expose online banking credentials and compromise confidentiality. Unauthorized changes to university examination results can compromise integrity. A ransomware attack that prevents hospital staff from accessing important records can compromise availability.

The CIA Triad also provides a foundation for understanding cybersecurity threats, vulnerabilities, and common attack methods. By identifying what needs protection and how it could be compromised, individuals and organizations can make more informed decisions about security controls and risk management.

For aspiring cybersecurity professionals, learning the CIA Triad is an important first step toward understanding how to protect computer networks, applications, and information systems. These principles provide a framework for approaching security challenges systematically rather than treating cybersecurity as a collection of unrelated tools and techniques.

Key takeaway: Effective cybersecurity requires more than preventing unauthorized access. It requires protecting sensitive information from disclosure, maintaining the accuracy and trustworthiness of data, and ensuring that essential systems and services remain accessible to authorized users.